

Taichi Kamei

+1 (236) 868-0229 | t.kamei0716@gmail.com | linkedin.com/in/taichikamei | github.com/Taichi-Kamei

EDUCATION

3rd Year Engineering Physics

University of British Columbia | Vancouver

Expected Graduation - May 2028

Relevant Courses: Solid and Fluid Mechanics, Mechanical Designs, Electric Circuit Analysis, Signals and Systems, Thermodynamics, Statistical Mechanics, Quantum Mechanics

Dean's Honor List 2025

EXPERIENCE

Quantum Photonic Researcher

UBC Blusson Quantum Matter Institute

May 2026 - Present

PROJECTS

Drone Flight Controller

Feb 2026 - Present

- Designed a power board for 4S LiPo, integrating BMS IC and 5V/5A buck converter
- Sized charge and discharge FETs for the BMS IC based on anticipated continuous/pulse drain current, DS voltage, and on-resistance to achieve safe handling and minimize heat dissipation
- Designed a Sallen-Key low-pass filter with non-inverting amplifier for amplifying 5kHz signal while attenuating 25kHz PWM noise, simulated its functionality on NgSpice
- Prototyping a flight controller PCB, integrating ESP32-S3-Mini-1, BMS, IMU, and a barometer
- Designing and fabricating a drone frame and a test rig for motor PID tuning

Autonomous Clue Detecting Robot

November 2025

- Developed an autonomous vehicle in Gazebo and ROS 1 using OpenCV for PID control driving, obstruction avoidance, and CNN clue detection
- Designed and implemented a finite state machine (FSM) capable of transitioning its states for moving obstruction detection, clue board detection, and drive under various track surface conditions
- Optimized a PID algorithm for precise center-lane driving using dual side lines instead of a typical single line-following, overcoming camera center offset from the geometric center of the lane

Autonomous Payload Retrieving Robot

June - August 2025

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

SKILLS

Software	Python, Java, C/C++, Assembly, VHDL, Linux, Git
Electrical	KiCAD, LTSpice, Soldering, Oscilloscope, DMM, Logic Analyzer
Mechanical	Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper
Embedded	ESP-IDF, Arduino, Raspberry Pi, FPGA, ROS, I2C, SPI